Student Research Internship

IoT & Sensor Systems Engineer

AquaSentinel AI | River Sensor Deployment in the Swiss Field

About the Project

AquaSentinel AI is an open-source research project building an early warning system for Swiss fisheries managers. Our AI model predicts dissolved oxygen (DO) — the variable that kills fish during summer heatwaves — across all Swiss rivers and lakes. To validate the model and create the first real-time DO data stream, we need a physical sensor station deployed at a pilot river site.

The project is funded by the Prototype Fund Switzerland (CHF 50,000) and runs November 2026 – February 2027. You will be based partly at OST Rapperswil — chosen specifically because the campus sits on Lake Zurich, giving you immediate access to a water body for sensor testing before any field installation.

Your Role

You will design, procure, calibrate, and deploy a real-world water quality sensor station at a Swiss river site defined in collaboration with Patrick Hager of Fischereiverein See + Gaster — a fishery manager who currently makes fish rescue decisions with only 24 hours' notice. Your sensor will give him 14 days.

Work is structured in four monthly milestones:

- **Month 1 —** Research & procurement: Survey sensor technologies used by Swiss monitoring agencies (BAFU, cantonal offices, Eawag). Conduct a market analysis of available DO, temperature, and water level sensors. Select and order hardware. Begin bench testing on campus (Lake Zurich access).
- **Month 2 —** Calibration & site preparation: Calibrate sensors against known reference measurements. Confirm field installation site with Patrick Hager. Prepare mounting, weatherproofing, and telemetry setup (GSM/LoRa).
- **Month 3 —** Field deployment: Install sensor station at the pilot river site. Commission real-time data transmission to the shared pipeline. Deliver live data stream to the AI team.
- **Month 4 —** Reserve & documentation: Troubleshooting, system hardening, sensor maintenance guide, open-source documentation.

What You Will Work With

- Sensor hardware: DO probes, temperature/depth loggers, telemetry modules (GSM/LoRa/NB-IoT)
- Reference networks: BAFU FOEN stations, cantonal monitoring systems, Eawag deployments
- Tools: Python for data ingestion, GitHub, field calibration equipment
- Location: OST Rapperswil (Lake Zurich campus) + pilot river site
- Team: weekly check-ins with project leads Olga and Tatyana; data handover to the ML team

What We Are Looking For

Required:

- Enrolled in a BSc or MSc program in Electrical Engineering, Computer Engineering, Environmental Engineering, IoT/Embedded Systems, or a related field
- Hands-on experience with microcontrollers, sensors, or embedded systems
- Interest in field work and working with real environmental data
- Comfortable working independently and in the field

Nice to have:

- Experience with water quality sensors or environmental monitoring hardware
- Knowledge of telemetry protocols (GSM, LoRa, NB-IoT, MQTT)
- Familiarity with Python data pipelines
- Interest in open-source hardware and responsible technology

What We Offer

- CHF 1,000/month stipend — 4-month duration (~15–20 hours/week)
- Academic credit: we will work with your university to support a thesis or project course connection
- OST Rapperswil campus access and Lake Zurich as a testing environment from day one
- Field deployment experience with real-world constraints — weather, permits, physical installation
- Your sensor data will power a live AI model and help protect Swiss fish populations

Logistics

- Duration: 4 months — November 2026 to February 2027
- Location: OST Rapperswil + pilot river site (Canton St. Gallen / Zurich area) + remote
- Start date: Flexible, ideally 1 November 2026
- Open to BSc and MSc students from any Swiss or international university
- OST students especially welcome — local campus access and potential academic credit alignment

How to Apply

Send a short email (one page maximum) to polar.bear.before.lunch@gmail.com with:

1. Your current program and year of study
2. Two or three sentences on why this project interests you
3. Any relevant hardware projects, lab work, or field experience

We will respond to all applications within one week.